

CSE320: Data Communications

Assignment 3 [Spring 2026] (Version 1.0)

Time Division Multiplexing (TDM)

1. A multiplexer combines four 100-kbps channels using a time slot of 2 bits. Show the output with four arbitrary inputs. What is the frame rate? What is the frame duration? What is the bit rate? What is the bit duration?
2. Five channels are multiplexed using TDM. Each channel can process 1000 bits per second. If 5 bits at a time are to be multiplexed using the concept of TDM with 2 synchronizing bits, then find the following.
 - (a) Input rate & input duration
 - (b) Frame size, frame rate & frame duration
 - (c) Output rate & output duration
 - (d) Output slot duration
3. Five channels are multiplexed using TDM. Each channel can process 1000 pages per 5 seconds. Each page has 20 lines, each line with 40 characters. If 5 lines at a time are to be multiplexed using the concept of TDM with 5 synchronizing bits, then find the following.
 - (a) Input rate & input duration
 - (b) Frame size, frame rate & frame duration
 - (c) Output rate & output duration
4. Six channels, three with a bit rate of 240 kbps and three with a bit rate of 120 kbps, are to be multiplexed. If 5 characters from each channel are to be multiplexed with three synchronization bits, then answer the following questions.
 - (a) Show how this can be achieved.
 - (b) What is the input rate and input duration?
 - (c) What will be the frame size, frame rate & frame duration?
 - (d) What will be the output rate & output duration?
 - (e) What will be the output slot duration?
5. Five channels, two with a bit rate of 240 kbps and three with a bit rate of 180 kbps, are to be multiplexed, with one synchronization bit and two interleaved bits. Answer the following questions.
 - (a) Show how this can be achieved.
 - (b) What is the input rate and input duration?
 - (c) What will be the frame size, frame rate & frame duration?
 - (d) What will be the output rate & output duration?
 - (e) What will be the output slot duration?
6. A control system has 8 devices (A-H). In the first time slot, only devices A, C, E, F, and H have data. In the second time slot, devices A, B, D, E, and G have data.
 - (a) What TMD technique should be used here (synchronous or statistical), and why?
 - (b) Sketch a figure of the TDM process and show the frames.
 - (c) Sketch a figure of the other TDM process and show the frames.

Error Detection & Correction

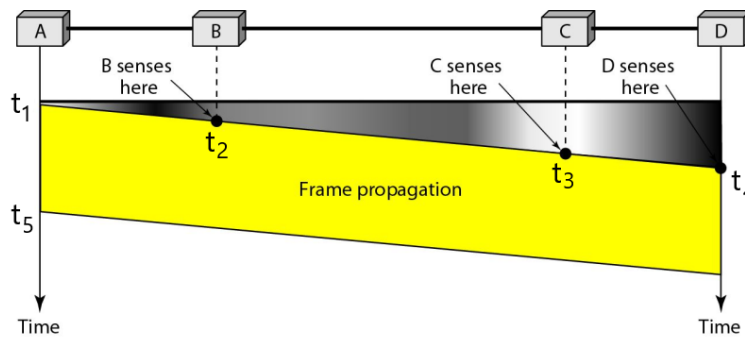
7. Consider the following coding scheme for error detection and correction.

Dataword	Codeword
00	00001111
01	01010011
10	10110100
11	11111001

- Write the scheme in $C(n, k)$ format. How many redundant bits are used in this scheme?
 - How many datawords and codewords are possible in this scheme? How many of the codewords are valid, and how many are invalid (extra)?
 - Find the maximum error detecting and correcting capability of this scheme.
 - Suppose the receiver receives two bit streams: 00011101 & 10111100. Find out whether they are correct or corrupted. If corrupted, find out whether it is possible to recover the original dataword.
8. Let **X** be the last 4 digits of your ID, and **D** and **G** be the binary representations of $X \% 256$, and $X \% 16$, respectively. Consider **D** as the dataword and **G** as the generator.
- Using **binary division**, find the codeword that is to be transmitted using CRC scheme.
 - Suppose an error occurs during transmission, and the **5th** bit from **MSB** is flipped. Can the CRC scheme detect this error? Show calculations.
9. Suppose you are the recipient of a data transmission system which uses CRC scheme for error detection and correction. In the following table we have four codewords received by you. The divisor/generator for each codeword is mentioned on the right column.

Codeword	Generator
100110100	1001
100110010	1001
100100110	1100
100101110	1110

- Write the scheme in $C(n, k)$ format. How many redundant bits are used in this scheme?
 - How many datawords and codewords are possible in this scheme? How many of the codewords are valid, and how many are invalid (extra)?
 - If your student ID is **even**, find out whether the **first** two codewords are correct or corrupted using **Polynomial division**.
 - If your student ID is **odd**, find out whether the **last** two codewords are correct or corrupted using **Polynomial division**.
10. Let us assume a packet is made only of four 16-bit words $(4501)_{16}$, $(1A4F)_{16}$, $(A309)_{16}$, $(B007)_{16}$. The given words are in hexadecimal values, where each digit can be represented by 4 bits. Remember that hexadecimal values range from 0000 – FFFF.
- Find out the checksum at the sender.
Hint: Proceed like this: First find out $4501 + 1A4F + A309 + B007$. If you cannot do it by hand, you can do it using a calculator [check out **this** youtube video on how to do it]. Then, check if the sum is more than 16 bits – if yes, we need to find out the wrap around sum to make it 16 bits. Then take 1's complement to negate it – this is the checksum at sender's end.
 - If the last hexadecimal digit in the second word is erred from F to A, will the receiver be able to detect it? Show calculations.
Hint: Find the sum of all received words and the checksum, wrap around if needed, and negate. If it equals to 0, the receiver will say NO ERROR. Else, error.



Multiple Access

11. In the following figure, $t_1 = 1.0\mu s$, $t_2 = 1.5\mu s$, $t_3 = 2.4\mu s$, $t_4 = 2.8\mu s$, $t_5 = 5.0\mu s$.
- Which multiple access protocol does this figure refer to?
 - Find the **transmission time** for this frame.
 - Find the maximum time the frame takes to travel between stations (**maximum propagation time**).
 - Find the vulnerable period in this setting, and compare it to the vulnerable periods of the **pure** and **slotted** versions of 'Just transmit it bro' protocol (aka ALOHA).
 - If the channel bandwidth is 10 Mbps, find the **minimum required frame size** in this setting.
12. A station continuously senses the medium and finds the following information:

Time interval (s)	Energy level (J)
0 - 4	0
4 - 10	5
10 - 14	10
14 - 20	0
20 - 25	5
25 - 30	20
30 - 40	0

- For each time interval, find the status of the medium (idle, transmission, collision).
 - Find out total time in which the medium was transmitting normally, and was in collision.
 - If a station wanted to send a frame, what time intervals would be available for it?
13. In a shared communication link, there are two stations on either side. They are using the CSMA/CD protocol to detect any collision occurring during transmission. The minimum frame size to detect collision is 1200 bytes, the speed of propagation is 5×10^7 m/s, and the bandwidth of the channel is 30 Mbps.
- Calculate the minimum transmission time to detect a collision.
 - Find out the vulnerable time of the system.
 - Identify the distance between the two stations.
14. In a smart city parking system, hundreds of parking sensors are installed across different parking slots. Each sensor sends availability updates to a central server using a shared communication channel. When a sensor has data to send, it first listens to the channel. If the channel is idle, the sensor transmits instantly. If the channel is busy, the sensor does not keep sensing the channel. Instead, it waits for a random time before sensing the channel again.
- Name which CSMA variant (1 persistent, non-persistent or p-persistent) is used here.
 - Discuss a major issue with the variant mentioned in (a).
 - CSMA protocol cannot detect collisions. Name the protocol that can detect a collision. Explain what happens after a collision is detected.